Arham Siddiqui

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Education

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering & Computer Science — *Expected May 2028*

- **Relevant Coursework:** Optimization Models in Engineering, Discrete Mathematics and Probability Theory, Machine Structures, Data Structures, The Structure and Interpretation of Computer Programs, Foundations of Signals Dynamical Systems and Information Processing, Linear Algebra and Differential Equations, Physics for Scientists and Engineers

Technical Skills

Languages: Python, C++, Java, TypeScript, JavaScript, HTML, CSS, SQL

ML/AI: PyTorch, Numpy, TensorFlow, Keras, scikit-learn, Pandas, Machine Learning Models, Deep Learning, Deep Learning Optimization, Neural Network Architectures, Reinforcement Learning, Multi-modal Learning, Foundation Models, Computer Vision, OpenCV, Supervised Learning, Unsupervised Learning, Transformers, LangChain, Large Language Models, Classification, ModelOps, AI Application, NLP, Contrastive Learning

Systems/Infrastructure: Object-Oriented Design, Data Structures, Algorithms, Software Design, Software Development, Software Optimization, Embedded Systems, Robotics, Neuroscience, Neurotechnology, Biotechnology, Cloud Infrastructure, Data Integration, Version Control Systems, Git, Jupyter, VSCode, React, Node.js, LaTeX, Arduino, Raspberry Pi, Slack, Debugging, Analytics

Experience

Apple | *Software Engineer (Contract)* — Jan 2026 – May 2026
Computer Vision, Data Pipelines, OpenCV

- Built an end-to-end computer vision pipeline to automatically validate large-scale image datasets for ML training.
- Designed multi-model image-quality scoring for blur, noise, content, and aesthetics, with backend aggregation logic for threshold-based filtering and human-in-the-loop review.

Google DeepMind | *Software Engineer (Contract)* — Sep 2025 – Mar 2026
Reinforcement Learning, AI Agents, Contextual Bandits, Bayesian Uncertainty

- Designed and built a Gemini-powered Google Calendar AI agent using LangChain, integrating tool-calling workflows for calendar reasoning, scheduling assistance, and adaptive task execution.
- Developed a Hybrid LinUCB contextual bandit to dynamically select agent autonomy levels, modeling user trust as a time-evolving latent state and using Bayesian-inspired uncertainty, confidence-bound exploration, and trust-aware rewards to balance personalization, safety, and long-term acceptance.

MIT Beaver Works Summer Institute | *Researcher* — Feb 2024 – Aug 2024
Multimodal ML, Signal Processing, NLP, Transformers

- Built four multimodal perception systems for assistive navigation across audio, vision, and language tasks.
- Implemented Discrete Fourier Transform-based feature extraction, sequential neural networks, NLP pipelines, and transformer-based models for real-world perception tasks.

Projects

Speech Foundation Models for Silent Speech Decoding | *Poster* — Sep 2025 – Present
Multimodal Deep Learning, Mechanistic Interpretability, PyTorch, Sensor Fusion, Contrastive Learning

- Building an interpretable silent-speech decoding framework extending a 30-class pipeline into a causal analysis system for speech self-supervised models.
- Implemented layer-wise probing, sparse autoencoder feature discovery, causal ablation, and activation steering on HuBERT/Wav2Vec2 hidden states to study articulatory and phonetic representations.
- Engineered multimodal encoders for lip landmarks, mouth video, UWB radar, mmWave/FMCW radar, and laser speckle signals using CNN-LSTMs, BiLSTMs, temporal attention, ResNet18, and 1D/2D convolutions.
- Improved speaker-independent generalization using supervised contrastive learning, domain-adversarial training, gradient reversal, and speaker-disjoint evaluation.

Overhead Movement Alarm Radar (OMAR) | *Patent* — Sep 2021 – Jun 2025
Embedded Systems, Computer Vision, Wearable Prototyping

- Built an assistive wearable to detect overhead obstacles for the blind using computer vision, ultrasonic sensors, Arduino Nano, and vibration motors; earned the California State Seal of Civic Engagement.
- Iteratively prototyped and tested with the same blind user over four years, improving detection range, feedback timing, sensor placement, and wearable form factor.

Machine Learning Research | *Audio Hazards, Rodent Behavior, Phishing ML, Defect Detection* — 2022 – Present

- Conducted long-term ML research across audio, vision, and language domains, developing skills in model design, literature review, experimental methodology, and comparative analysis.
- Built and compared CNN, LSTM, hybrid, ensemble, and classical ML models using engineered features such as MFCCs, spectral representations, image features, and structured behavioral/testing data.